Seminar V

IVP
$x(0) = \eta$

Def 1 [Dynamical systems vocabulary]

The flow of $\dot{x} = f(x)$ is the function $\varphi : D = \{(t,\eta) \in \mathbb{R}^2 : t \in I_\eta\} \to \mathbb{R}$, $(t,\eta) \mapsto \varphi(t,\eta)$.

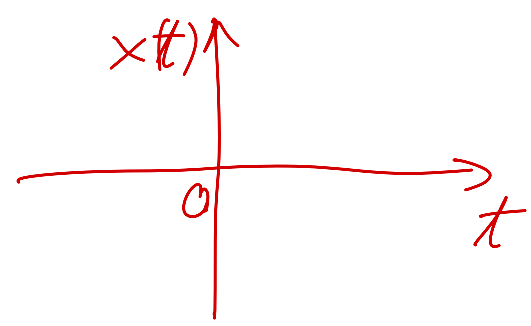
The value $\varphi(t,\eta)$ is said to be the state of the system at time t when starting with the initial state η.

The orbit corresponding to the initial state $\eta \in \mathbb{R}$ is:

$$\gamma_\eta = \{\varphi(t,\eta) : t \in I_\eta\} \subset \mathbb{R}$$

The phase portrait of $\dot{x} = f(x)$ is the representation on the line $\mathbb{R}$ of the orbits, togheder with an arrow on each orbit that indicates the future states.

2 D



1 D

Properties of the flow: (i) $\varphi(0,\eta) = \eta \quad \forall \eta \in \mathbb{R}$

(ii) $\varphi(t+s,\eta) = \varphi(t, \varphi(s,\eta)) \quad \forall \eta \in \mathbb{R}, s \in I_\eta, t \in I_{\varphi(s,\eta)}$

(iii) D is an open subset of $\mathbb{R}^2$ and φ is a C^1 function.

Procedure to represent the phase portrait:

Step 1: Find all the equilibrium points (or stody states) (solve the eq $f(x) = 0$ for $x \in \mathbb{R}$)

Step 2: Represent the equilibrium points on the line $\mathbb{R}$

Step 3: Det. the sign of f on each orbit. If the sign $+$, the arrow must indicate to the right, while if the sign $-$, the arrow must indicate to the left.

Def 2: An equilibrium solution η^* of $\dot{x} = f(x)$ is said to be an attractor when $\exists r > 0$ s.t.: $|\eta - \eta^*| < r$ we have $\lim_{t \to \infty} \varphi(t,\eta) = \eta^*$.

The basin of attraction of an attractor η^* is

$$A_{\eta^*} = \{ \eta \in \mathbb{R} : \lim_{t \to \infty} \varphi(t, \eta) = \eta^* \}$$

An equilibrium solution η^* of $\dot{x} = f(x)$ is said to be a repeller when $\exists\, r > 0$ s.t.: $|\eta - \eta^*| < r$ we have $\lim_{t \to -\infty} \varphi(t, \eta) = \eta^*$

Theorem 5: [The liniarization method]

Let η^* be an equilibrium solution of $\dot{x} = f(x)$.

If $f'(\eta^*) < 0$ then η^* is an attractor.

If $f'(\eta^*) > 0$ then η^* is a repeller.

Type and stability of linear planar systems

Consider $\dot{X} = A \cdot X$: (1), where $A \in M_2(\mathbb{R})$, $\det A \neq 0$.

Denote by $\lambda_1, \lambda_2 \in \mathbb{C}$ the two eigenvalues of A. We know that $\det A = \lambda_1 \cdot \lambda_2$

Remark: 1) $\det A \neq 0 \Leftrightarrow 0 \in \mathbb{R}^2$ is the only equilibrium point of (1)

(2) $\det A \neq 0 \Leftrightarrow \lambda_1 \neq 0$ and $\lambda_2 \neq 0$

Def 1: We say that the equilibrium $0 \in \mathbb{R}^2$ of (1) is a

NODE: $\lambda_1, \lambda_2 \in \mathbb{R}$ and either $\lambda_1 \leq \lambda_2 < 0$ or $0 < \lambda_1 \leq \lambda_2$

SADDLE: $\lambda_1, \lambda_2 \in \mathbb{R}$ and $\lambda_1 < 0 < \lambda_2$

FOCUS: $\lambda_{1,2} = \alpha \pm i\beta$, $\alpha \neq 0$, $\beta \neq 0$

CENTER: $\lambda_{1,2} = \pm i\beta$, $\beta \neq 0$

Theorem 1: If $Re(\lambda_1) < 0$ and $Re(\lambda_2) < 0$ then the equilibrium $0 \in \mathbb{R}^2$ of $\dot{X} = A \cdot X$ is a global attractor.

If $Re(\lambda_1) > 0$ and $Re(\lambda_2) > 0$ then the equilibrium $0 = (0,0) \in \mathbb{R}^2$ of $\dot{X} = A \cdot X$ is global repeller.

Def 2: Let $f \in C^1$ and $\eta^* \in \mathbb{R}^2$ s.t. $f(\eta^*) = 0$. We say that the equilibrium point η^* of $\dot{x} = f(x)$ is stable if $\forall\, \varepsilon > 0$, $\exists\, \delta > 0$ s.t. $\|\eta - \eta^*\| < \delta$ we have $\|\varphi(t, \eta) - \eta^*\| < \varepsilon$, $\forall\, t \in (0, \infty)$. We say that it is unstable if it is not stable.

Theorem 2: Any CENTER is stable.
Any SADDLE is unstable.